

Bharath Velamala

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Technical Skills

Languages: Python, Java, SQL, R, Rust, Shell Scripting
Data & Pipelines: Apache Beam, Dataflow, Spark, Airflow, dbt, Kafka, Pub/Sub, ETL/ELT, Dimensional Modeling
AI/LLM: HuggingFace, PyTorch, RAG, Vector Embeddings, Gemini/LLM integration, Vertex AI, OCR
Databases: PostgreSQL, BigQuery, Snowflake, NoSQL (Firestore, MongoDB), Redis, FHIR Store
Cloud & Infra: GCP (Dataflow, BigQuery, Cloud Run, Dataproc, GCS, GKE, Pub/Sub), Terraform, Docker, CI/CD
Certifications: GCP Professional Data Engineer, GCP Associate Cloud Engineer

Professional Experience

Data Engineer II – Centauri Health Solutions, Tempe, AZ **Jul 2025 – Present**

- Architected **CLARION**, a multi-tenant clinical data platform; designed **end-to-end ETL pipelines** with schema contracts and SCD Type 2 tracking processing **1M+ weekly records**, enforcing data quality SLAs for **5+ downstream consuming teams**.
- Built a **streaming pipeline** ingesting XML (CDA) and ZIP/PDF documents into FHIR bundles on GCP Dataflow; resolved worker OOM and timeout errors via per-page parallelism and file-handle lifecycle management, cutting failures **~80%**.
- Partnered with clinical stakeholders to design a **Map-Reduce NER inference pipeline** on Dataflow across **16K documents (4.4M pages)**; hybrid rule-based + LLM classification populating BigQuery datasets for AI-assisted medical coding.
- Built **MVPs and UI validators** enabling non-technical stakeholders to review ML extraction outputs before production promotion, directly accelerating sign-off cycles.

Data Science Intern (Biostatistics) – Mayo Clinic, Rochester, MN **May 2024 – May 2025**

- Co-authored and published **medspaCyV (PMC12374723)**, an open-source clinical NLP tool; custom sentence tokenizer, sectionizer, and NER visualizer converting unstructured EHR notes into **structured, queryable data assets** for clinical decision-making.
- Engineered an **event-driven NLP microservice** on GCP (Cloud Run, Firestore) translating clinical notes into structured assets; async concurrency and optimized service boundaries reduced pipeline latency **60%**.

Data Engineer – HSBC, Hyderabad, India **Jan 2023 – Aug 2023**

- Migrated **2M+ weekly financial records** from on-premise mainframe to GCP (Airflow, Dataflow, BigQuery), reducing execution time **35%**; established regression testing suites to safeguard data integrity across deployments.
- Designed a **multi-tenant data model** (PostgreSQL + BigQuery) with clustering, partitioning, and schema validation across **20+ critical tables**; partnered with **3+ upstream/downstream teams** to achieve **100% schema compliance** post-migration.
- Built **distributed Spark/Dataproc pipelines** for financial analytics; applied windowed aggregations and optimized joins to cut processing time **40%** and cloud costs **25%**.

Associate Data Engineer – SpringML Inc., Hyderabad, India **Jan 2021 – Dec 2022**

- Engineered a **large-scale Apache Beam pipeline (Java)** on Dataflow, processing **1.5M+ weekly records**; tuned autoscaling, sharding, and PII masking for **8-min end-to-end latency (50% improvement)** with full data governance compliance.
- Designed **dimensional data models (STAR schema)** in BigQuery for Looker/Data Studio reporting; partnered with business stakeholders across **3+ projects** to translate requirements into materialized views and analytics-ready datasets.
- Implemented **fault-tolerant CDC pipelines** (BigQuery, Dataflow, Airflow + dbt SQL MERGE) for delta processing; Airflow sensors for upstream file arrival accelerated data availability **35%** for downstream stakeholders.
- Launched a **Cloud Run microservice** orchestrating Document AI via Pub/Sub with automated schema validation; processed **100K+ claim uploads** and cut manual verification latency **70%**.
- Authored **Terraform IaC + Cloud Build CI/CD** across dev/staging/prod with approval gates and automated rollback; reduced release lead time **40%** with zero manual intervention.

Projects

Agentic Scratchpad Scheduler | Python, Gemini, NetworkX, Multi-Agent **MLSys 2026 — 2nd Place, Track B**

- Neurosymbolic **multi-agent pipeline** for tensor DAG scheduling on memory-limited hardware; LLM proposes op fusion, deterministic solver enforces constraints; **7.77× aggregate speedup** over baseline.

GCP Dataflow Template MCP Server | Python, MCP, Cloud Run, Apache Beam, Docker

- Built and **deployed an MCP server** (Cloud Run) exposing Dataflow pipeline scaffolding as agent-callable tools; enables AI assistants to scaffold and deploy standardized Beam pipelines, reducing bootstrap time from hours to minutes.

WAL Distributed | Rust, Tokio, gRPC/tonic, Raft, Prometheus

- Write-Ahead Log + full Raft consensus**; crash-safe segment engine at **429 MiB/s, 46ns** encode; leader election, log compaction, InstallSnapshot RPC; Prometheus metrics.

Education

University of Arizona **Aug 2023 – May 2025**

Master of Science, Data Science | GPA: 4.0/4.0

Coursework: Data Mining, Machine Learning, NLP, Big Data, Deep Learning, AI (Teaching Assistant)

Indian Institute of Technology Patna **Aug 2017 – May 2021**

Bachelor of Technology, Mechanical Engineering | GPA: 3.5/4.0